GHOST STORY LENS

Product Requirements Document

An Explainable, Corpus-Grounded AI Genre Analyzer

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Product premise

Genre classification is an interpretive argument supported by textual evidence, not the discovery of an objectively fixed property.

Document Control

Field	Specification
Purpose	Define product, design, AI, data, export, accessibility, validation, and implementation requirements for the first production version of Ghost Story Lens.
Primary audience	Product owner, Replit developer(s), UX designer, AI engineer, research annotators, testers, and dissertation readers reviewing the project methodology.
Decision authority	Jong-Keyong (Jake) Kim serves as product owner and approves changes to the default rubric, corpus, visual system, and public-facing explanatory language.
Source of requirements	The revised Ghost Story Lens product blueprint developed in this project, including the interactive Relational Manifestation Map and combined PDF export requirement.
Research grounding	The dissertation's relational model of spectral sources, ambient mediators, and attuned perceivers, operationalized here as the ghost–mediator–seer network.

Revision History

Version	Date	Author	Change
0.1	18 July 2026	Jong-Keyong (Jake) Kim / ChatGPT	Initial concept: corpus-grounded ghost-story spectrum and explainable evidence.
0.2	18 July 2026	Jong-Keyong (Jake) Kim / ChatGPT	Added interactive Relational Manifestation Map and visual node specifications.
0.3	18 July 2026	Jong-Keyong (Jake) Kim / ChatGPT	Added restrained non-stereotypical design system, About page, multilingual output, and static exports.
1.0	18 July 2026	Jong-Keyong (Jake) Kim / ChatGPT	Converted the approved blueprint into an implementation-oriented PRD; combined analysis and map export changed from JPG to PDF.

Version	Date	Author	Change
1.1	19 July 2026	Jong-Keyong (Jake) Kim / Claude	Replaced the restrained-luminous visual system with a dual-axis analytical/spectral design and a will-o'-the-wisp cursor motif; kept body text fixed at 13 pt and switched reading width from a fixed column to fluid; restructured primary navigation (Home, About, Sample Analysis, Analyze, My Analysis) and rerouted the Home page to About; added a My Analysis page and expanded About-page usage guidance for multilingual output, the Relational Manifestation Map, and the AI research conversation.
1.2	19 July 2026	Jong-Keyong (Jake) Kim / Claude	Simplified export to a single combined PDF report (analysis summary and relational map together); removed the separate Analysis Summary JPG and Relational Map JPG exports and renamed the action from Download Results to Download Result.
1.3	19 July 2026	Jong-Keyong (Jake) Kim / Claude	Added a Light/Dark Mode theme toggle (new 10.7) with a Gothic lancet-arch icon in the top-right of the primary navigation; added the Dark Mode palette table to 10.3; extended the accessibility contrast requirement to both modes; the combined PDF export always uses the Light Mode palette regardless of on-screen theme.
1.4	19 July 2026	Jong-Keyong (Jake) Kim / Claude	Committed Custom Rubric Mode to this MVP (FR-022 raised from Should to Must; resolved the corresponding open decision in 21.2); added a new 8.2.1 Custom Rubric Editor subsection (FR-024 through FR-028) specifying the account gate, criterion fields, weight validation, score-band editing, and exclusions; clarified that 8.7's approval/versioning/attribution flow covers both AI-proposed and user-authored rubric changes; added Release Acceptance Criterion 19.

Contents

1. Executive Summary

Ghost Story Lens is an explainable, corpus-grounded web application that evaluates a literary text against a transparent and revisable definition of the ghost-story genre. A user uploads a supported file or pastes a story, and the application returns a 1–10 Ghost-Story Fit score, a separate analytical-confidence assessment, criterion-level explanations, supporting and contradicting passages, supernatural-entity classification, corpus comparisons, rubric-gap findings, an interactive Relational Manifestation Map, and an AI research conversation.

The product is not a binary detector and must not present its score as a statistical probability that a text “is” a ghost story. The score measures correspondence with a specified rubric and corpus calibration. The result is therefore a documented interpretive argument that users can inspect, question, revise, and export.

The first production version will be developed and deployed through Replit. GPT-5.6 Sol will perform structured evidence extraction, criterion evaluation, entity classification, network construction, multilingual explanation, and research dialogue. The application's own rule-based logic (not the AI model) will calculate weighted scores, assign score bands and node sizes, validate references, preserve versions, and generate static exports.

Core product promise

Ghost Story Lens shows where a story falls on a ghost-story spectrum, why it falls there, how the haunting is relationally mediated, who perceives it, and whether the story exposes a limitation in the current definition.

1.1 Product Name and Positioning

Element	Approved Language
Product name	Ghost Story Lens
Tagline	How ghostly is this story—and why?
Descriptor	An Explainable AI Genre Analyzer
Primary value	Transparent, evidence-linked, relational literary classification.
Differentiator	The rubric can be questioned and revised when a submitted story produces meaningful evidence that the current criteria cannot represent.

1.2 Product Description

Ghost Story Lens is an explainable, corpus-grounded literary analysis platform that places texts on a revisable ghost-story spectrum. It identifies the evidence behind its classification, distinguishes ghosts from adjacent supernatural forms, creates an interactive map of manifestation and perception, and allows users to challenge both the analysis and its underlying criteria through dialogue with AI.

2. Product Context and Problem Statement

2.1 Problem

Ghost-story classification is often handled through unarticulated intuition, simple keyword matching, anthology precedent, or an overly rigid requirement that a visible dead person appear. These approaches obscure the historical and relational complexity of spectral narrative. They also collapse ghosts, demons, vampires, doubles, fairies, hallucinations, fraudulent hauntings, metaphorical specters, and occult agencies into one undifferentiated supernatural category.

Existing general-purpose AI systems can produce persuasive classifications, but their decisions are difficult to reproduce when the criteria, evidence, comparison set, weighting, and model version remain hidden. Researchers and students therefore need an application that makes the basis of classification inspectable and that treats disagreement as part of the method rather than as an error to suppress.

2.2 Research Opportunity

The application operationalizes a literary-research question: how does a ghost become narratively perceptible through relations among a spectral source, material or immaterial mediators, and one or more attuned perceivers? It converts that question into a practical workflow for corpus adjudication, close reading, teaching, and methodological reflection.

- Move from a yes-or-no classification to a graded, evidence-based spectrum.
- Separate genre fit from confidence and from supernatural-entity type.
- Represent haunting as a relational configuration rather than an isolated apparition.
- Treat rubric failure and model resistance as potentially meaningful findings.
- Create a reproducible record of evidence, scores, prompts, models, and user revisions.

2.3 Product Hypothesis

Hypothesis

Users will make more defensible and more nuanced literary classifications when the application shows both supporting and contradicting passages, separates entity ontology from genre fit, and visualizes the ghost–mediator–seer relation as an evidence-linked network.

3. Vision, Goals, and Non-Goals

3.1 Vision

Ghost Story Lens should become an accessible digital-humanities environment in which literary genre is examined as a historically situated, relational, and revisable system. The application should help a first-year undergraduate understand why a classification was made while also giving a literary scholar enough evidence, metadata, and methodological documentation to challenge or reuse the analysis.

3.2 Product Goals

1. Provide an interpretable 1–10 Ghost-Story Fit score tied to a visible and versioned rubric.
2. Show passage-level evidence for and against every major classification claim.

3. Differentiate human ghosts from adjacent supernatural beings, psychological phenomena, fraud, and metaphorical spectrality.
4. Create an interactive Relational Manifestation Map that models how the ghost manifests, is mediated, and is perceived.
5. Allow users to question classifications, network relations, and rubric limitations through an AI research conversation.
6. Preserve user-approved revisions without erasing the original model output.
7. Support multilingual interfaces and results while preserving evidence in the source language.
8. Generate a single combined PDF research report integrating the analysis summary and network map.
9. Document model, prompt, rubric, corpus, and analysis versions for reproducibility.
10. Present a visual identity that stages a legible tension between analytical structure and spectral suggestion, expressed symbolically rather than through literal Gothic or horror imagery.

3.3 Non-Goals for the MVP

- Proving a universal definition of the ghost story across all historical and cultural traditions.
- Replacing literary interpretation, human annotation, or expert judgment.
- Providing a legal determination about copyright, fair use, or publication rights.
- Performing high-quality OCR on image-only or degraded scans.
- Analyzing an entire user corpus in one batch.
- Allowing unrestricted public redistribution of uploaded copyrighted texts.
- Automatically changing the default rubric or research corpus without product-owner approval.
- Fine-tuning a model in the first release.
- Serving as a horror-entertainment or game application.

4. Target Users and Use Cases

4.1 Primary Personas

Persona	Needs and Uses
Undergraduate Reader	Needs accessible explanations, definitions, visible passages, and a clear account of how genre claims are supported. Uses Reader View and Sample Analysis.
Graduate Researcher	Tests difficult cases, compares stories, develops working definitions, and exports evidence for seminar papers, theses, or dissertation work.
Literary Scholar	Uses Research View, rubric versions, network corrections, corpus comparisons, and structured exports to support research design and close reading.
Instructor	Demonstrates how definitions are constructed, assigns comparative analyses, and asks students to contest AI-generated evidence and relations.

Persona	Needs and Uses
Corpus Curator	Screens candidate texts, documents inclusion or exclusion decisions, and identifies cases that require manual adjudication.

4.2 Representative User Stories

ID	User Story
US-01	As a student, I want to paste a story and receive a clear explanation of where it falls on the ghost-story spectrum so that I can understand the genre decision.
US-02	As a researcher, I want every score and graph relation to link to passages so that I can verify the analysis against the primary text.
US-03	As a corpus curator, I want the application to distinguish a demon from a human ghost so that keyword overlap does not produce a false inclusion.
US-04	As a scholar, I want to see how manifestation is mediated through objects, spaces, sounds, documents, or technologies so that the network can support a relational interpretation.
US-05	As an instructor, I want a simplified Reader View and a Sample Analysis so that students can learn the method before uploading a text.
US-06	As a user working in Korean, I want the interface and report in Korean while quotations remain in the original language so that the evidence is not erased by translation.
US-07	As a researcher, I want to dispute a node or edge without deleting the original model output so that interpretive revision remains auditable.
US-08	As a user, I want to download a combined PDF containing the summary and network map so that I can review or share a stable research record.

5. Product Principles

Principle	Product Consequence
Interpretive, not absolute	The product reports fit with a named rubric, not an objective probability that a text belongs to a natural category.
Evidence before fluency	A persuasive explanation is insufficient unless each important claim is anchored to a retrievable passage.
Fit, confidence, and ontology are separate	A story may fit the rubric strongly while remaining ontologically ambiguous, or fail the ghost-story rubric with high confidence because its entity is demonic.
Relationality over isolated entities	The map models manifestation pathways among ghost, mediators, and seers rather than merely counting characters or supernatural words.

Principle	Product Consequence
Counterevidence is mandatory	Every report searches for the strongest evidence against the initial classification.
Rubrics are versioned arguments	Criteria may change, but no default change occurs silently or retroactively.
Human correction is first-class data	User disagreement and revision are preserved as part of the analytical record.
Cultural scope is explicit	The default rubric is historically situated and must not be presented as culturally universal.
Design expresses the argument	The interface stages a visible tension between an analytical/structural register and a spectral/suggestive register (e.g. a cursor-following will-o'-the-wisp), rather than relying on literal Gothic or horror iconography.

6. Scope and Release Strategy

6.1 MVP Scope

- Single-story analysis through pasted text, TXT, DOCX, and text-based PDF.
- Default British Ghost Story Criteria rubric, visibly versioned.
- Corpus-Calibrated, Comparative, and Custom Rubric modes. Custom-rubric editing is limited to signed-in, authorized accounts and ships in this MVP.
- Ghost-Story Fit score, category, confidence, criterion breakdown, entity classification, and evidence panels.
- Interactive Relational Manifestation Map with evidence-linked nodes and edges.
- Three to five corpus comparisons.
- Rubric-gap detection and proposed revisions.
- Context-aware AI research conversation.
- Multilingual interface and report generation in the approved initial languages.
- A single combined PDF export integrating the analysis summary and network map.
- About page, Sample Analysis, privacy statement, and creator/contact information.
- Versioned storage for analyses, revisions, prompts, models, rubrics, and corpora.
- Light/Dark Mode theme toggle with a restrained, Gothic-arch icon in the primary navigation.

6.2 Deferred Scope

- Batch corpus ingestion and analysis.
- Collaborative annotation teams and instructor dashboards.
- Automated OCR and image-heavy PDF analysis.
- User-uploaded reference corpora.
- Public rubric marketplace or community sharing.
- Institutional single sign-on and advanced role management.

- Model fine-tuning.
- Native mobile applications.
- Automatic publication-ready MLA citation generation for uploaded primary texts.

6.3 Proposed Release Phases

Phase	Deliverable
Phase 0: Prototype	Demonstrate upload, structured analysis, spectrum display, and a non-editable sample network on a small benchmark.
Phase 1: MVP	Deliver complete single-story workflow, interactive network, evidence, AI conversation, multilingual output, and required exports.
Phase 2: Research Expansion	Add editable custom rubrics, larger comparison sets, user annotations, advanced network filters, and structured research exports.
Phase 3: Teaching and Collaboration	Add classroom spaces, assignments, annotation comparison, collaborative adjudication, and instructor controls.

7. End-to-End User Journey

1. The user opens the Home page, reads the concise product statement, or opens Sample Analysis.
2. The user selects Analyze, pastes a story or uploads a supported file, and optionally enters metadata.
3. The application detects language and extraction quality, assigns stable passage identifiers, and asks for correction only when needed.
4. The user selects Corpus-Calibrated, Comparative, or Custom Rubric mode and chooses interface and output languages.
5. The application runs evidence extraction, entity classification, criterion evaluation, counterevidence analysis, network construction, corpus comparison, and rubric-gap analysis.
6. The application's own rule-based logic validates organized outputs, calculates the weighted fit score, assigns classification and confidence labels, and generates the results page.
7. The user reviews summary, spectrum, evidence, criterion breakdown, interactive network, ontology analysis, comparisons, and rubric gaps.
8. The user asks questions or proposes changes through the AI research conversation and may approve revisions to the analysis or personal/custom rubric.
9. The user downloads the combined analysis-and-map report as PDF.
10. The system stores or discards the text and analysis according to the user’s privacy selection.

8. Functional Requirements

8.1 Input and Text Preparation

ID	Requirement	Specification	Priority

ID	Requirement	Specification	Priority
FR-001	Paste input	Provide a large plain-text editor that accepts a complete story and preserves paragraph breaks.	Must
FR-002	File upload	Accept TXT, DOCX, and text-based PDF files. Validate MIME type, extension, size, and extraction success.	Must
FR-003	Image-only PDF handling	Detect image-only or severely degraded PDFs and decline analysis with a clear explanation.	Must
FR-004	Metadata	Allow optional title, author, publication date, language, source, and user notes.	Must
FR-005	Language detection	Detect source language and allow manual correction before analysis.	Must
FR-006	Passage indexing	Assign stable identifiers to paragraphs or passage windows so all evidence can be retrieved and exported.	Must
FR-007	Completeness warning	Estimate whether the text appears incomplete and display a warning that may lower confidence.	Should
FR-008	Privacy choice	Before analysis, let users choose whether the submitted text and report may be saved.	Must

8.2 Analytical Modes

ID	Requirement	Specification	Priority
FR-020	Corpus-Calibrated Mode	Apply the default rubric and comparison corpus; display "Calibrated on British ghost fiction, 1764–1913."	Must
FR-021	Comparative Mode	Apply the rubric to other periods, languages, and traditions and display an exploratory-analysis qualification.	Must
FR-022	Custom Rubric Mode	Allow authorized, signed-in users to adjust criteria, definitions, weights, exclusions, and score boundaries without overwriting the default rubric. See FR-024 through FR-028 for editor-level detail.	Must
FR-023	Rubric selection	Show rubric name, version, calibration, and last-modified date before analysis.	Must

8.2.1 Custom Rubric Editor

The editor is available only inside Research View to a signed-in, authorized account (User.role, per 14.1); an unauthenticated guest sees the Custom Rubric mode card but cannot open the editor, and selecting it prompts sign-in instead. A saved edit follows the same proposal, approval, versioning, and attribution flow as an AI-proposed rubric-gap revision (see 8.7); the only difference is who authored the change.

ID	Requirement	Specification	Priority
FR-024	Account gate	Require a signed-in account before the editor opens; guests are redirected to sign-in and their in-progress analysis is preserved.	Must
FR-025	Criterion fields	Allow adding, removing, renaming, and redefining criteria; each criterion has a name, a definition (free text), and a weight. Removing a criterion below the minimum of three is blocked.	Must
FR-026	Weight validation	Block saving unless all criterion weights sum to exactly 100%; show the running total and the remainder live as the user edits.	Must
FR-027	Score band editing	Allow editing the five score-band boundaries (Appendix A) subject to monotonic, non-overlapping ranges spanning exactly 1.0 to 10.0; reject any gap or overlap between adjacent bands.	Should
FR-028	Exclusions	Allow defining a condition (e.g. a confirmed-fraud entity classification) that caps or floors the overall fit score regardless of the weighted total, with the applied exclusion shown transparently on the results page.	Should

8.3 Analysis Pipeline

ID	Requirement	Specification	Priority
FR-030	Evidence extraction	Identify apparent ghosts, deceased persons, supernatural entities, seers, mediators, locations, objects, sensory manifestations, explanations, and key passages.	Must
FR-031	Criterion scoring	Return criterion scores, evidence, counterevidence, rationale, and confidence in validated organized output.	Must
FR-032	Counterevidence pass	Run a separate pass seeking evidence that undermines the initial classification or indicates another entity type.	Must
FR-033	Network extraction	Return nodes, edges, roles, relation types, importance, confidence, and supporting passages.	Must
FR-034	Corpus comparison	Identify three to five relevant comparison texts using relational, ontological, thematic, and semantic similarity.	Must
FR-035	Rubric-gap pass	Identify important evidence that no current criterion adequately represents and propose bounded revisions.	Must
FR-036	Application-Side Calculation	Application code, not the model, computes weighted scores, bands, node-size tiers, and export metadata.	Must

ID	Requirement	Specification	Priority
FR-037	Citation validation	Reject or flag model evidence references that cannot be matched to stored passages.	Must

8.4 Scoring and Classification

ID	Requirement	Specification	Priority
FR-040	Fit score	Calculate a decimal Ghost-Story Fit score from 1.0 to 10.0 from criterion scores and versioned weights.	Must
FR-041	Score bands	Map the score to Not a Ghost Story, Spectral-Adjacent Narrative, Borderline or Ambiguous Ghost Story, Strong Ghost Story, or Paradigmatic Ghost Story.	Must
FR-042	Confidence	Report analytical confidence separately from genre fit.	Must
FR-043	Entity type	Report primary and optional secondary entity classifications independently of fit score.	Must
FR-044	Version display	Display rubric, corpus, model, prompt, and analysis versions in Research View and exports.	Must
FR-045	No probability language	Do not label the score as a probability, certainty percentage, or objective genre membership.	Must

8.5 Results Page

ID	Requirement	Specification	Priority
FR-050	Result header	Show title, author, score, band, confidence, primary entity, rubric, and calibration.	Must
FR-051	Spectrum	Show a 1–10 mist-to-navy spectrum with an emerald marker and accessible text equivalent.	Must
FR-052	Interpretive summary	Provide a concise paragraph stating the principal classification and complications.	Must
FR-053	Why It Fits	Show three to five major supporting claims with expandable passages.	Must
FR-054	Why It May Not Fit	Show strongest counterevidence and alternative explanations.	Must
FR-055	Criterion breakdown	Provide score, weight, evidence, counterevidence, confidence, and explanation for every criterion.	Must
FR-056	Ontology analysis	Explain whether the entity is a ghost, demon, double, hallucination, fraud, or adjacent phenomenon.	Must

ID	Requirement	Specification	Priority
FR-057	Corpus comparisons	Show similarities and differences for three to five reference texts.	Must
FR-058	Rubric gaps	Show uncovered evidence and proposed criterion revisions.	Must
FR-059	View modes	Provide Reader View and Research View.	Must

8.6 AI Research Conversation

ID	Requirement	Specification	Priority
FR-060	Context	The assistant may access the submitted text, evidence, scores, entity labels, network, comparisons, rubric, and revision history.	Must
FR-061	Response types	Label responses as Explanation, Alternative Reading, or Proposed Revision.	Must
FR-062	Evidence grounding	Claims about the story must link to stored passage identifiers.	Must
FR-063	No silent changes	The assistant may not change stored results, the network, or a rubric without explicit user approval.	Must
FR-064	Network questions	Support questions about node importance, seer status, missing mediators, and relation direction.	Must
FR-065	Rubric impact preview	Before an approved rubric change, show expected impact on the submitted story and benchmark cases when available.	Should

8.7 Rubric Revision and Audit Trail

This proposal, approval, versioning, and attribution flow applies uniformly regardless of who authored the change: an AI-proposed rubric-gap revision (8.6) and a user-authored edit made in the Custom Rubric Editor (8.2.1) both become a Gap Proposal object, both require explicit approval, and both create a new rubric version rather than overwriting the default.

ID	Requirement	Specification	Priority
FR-070	Gap proposal	Create a proposal containing current wording, proposed wording, motivating evidence, rationale, risks, and projected impact.	Must
FR-071	Approval	Require explicit approval before saving a revision.	Must
FR-072	Versioning	Create a new rubric version rather than overwriting the previous version.	Must
FR-073	Attribution	Store approving user, date, model, prompt, passages, and affected texts.	Must

ID	Requirement	Specification	Priority
FR-074	Reanalysis	Allow users to rerun or compare an analysis under two rubric versions.	Should

8.8 My Analysis Page

ID	Requirement	Specification	Priority
FR-080	Analysis list	Show the authenticated user's saved analyses with title, date, rubric version, and score.	Must
FR-081	Reopen	Selecting an entry reopens the full results page for that analysis.	Must
FR-082	Rerun and compare	From an entry, allow the user to rerun the analysis under the current rubric version or compare it against a prior rubric version (see FR-074).	Should
FR-083	Deletion	Allow the user to delete a saved analysis and its exports, consistent with PRI-003.	Must
FR-084	Export re-access	Allow the user to re-download a previously generated combined PDF report for a saved analysis without regenerating it.	Should

9. Relational Manifestation Map

The Relational Manifestation Map is a signature product feature. It must visualize how a spectral presence becomes narratively perceptible, not merely which characters and objects co-occur. The network represents pathways among the ghost or apparent entity, the material and immaterial mediators of manifestation, and the ghost-seer or ghost-seers who perceive, interpret, record, or receive the phenomenon.

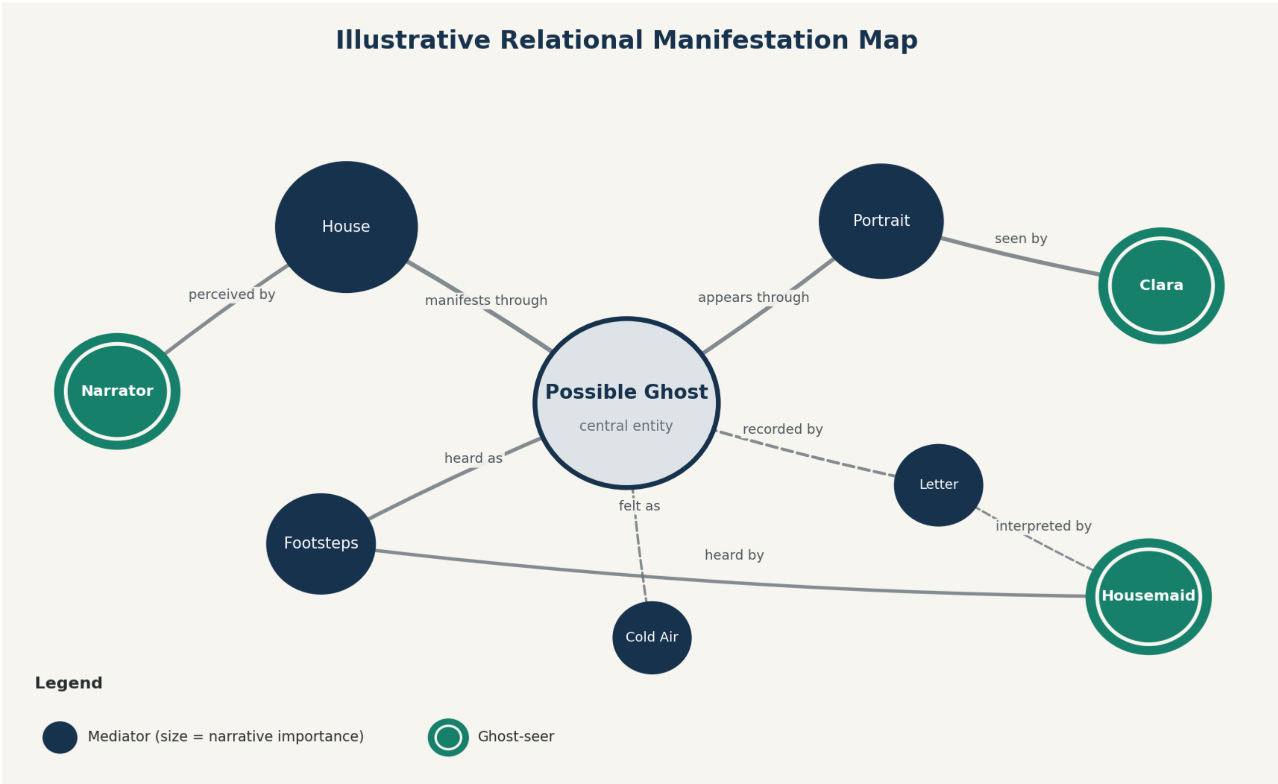

Figure 1. Illustrative network design. Final layouts and labels must be generated from evidence in the submitted story.

9.1 Node Requirements

Node Type	Required Representation
Ghost / apparent entity	Central by default; mist-gray or pearl interior; deep-navy outline; labeled with a confirmed or qualified identity. Multiple entities may appear when the text clearly differentiates them.
Mediator	Deep navy, single-ring node. Represents an object, space, body, sensory condition, atmosphere, document, technology, or human intermediary through which manifestation becomes perceptible.
Ghost-seer	Emerald, double-ring node containing a proper name or stable narratological role such as Unnamed Narrator or Housemaid.
Dual-role character	Retains the ghost-seer visual treatment and receives a role badge indicating an additional mediating or interpreting function.

9.2 Mediator Size Tiers

Tier	Meaning
Large	Primary mediator: causally central, structurally necessary, recurrent across major scenes, or decisive at the climax or resolution.

Tier	Meaning
Medium	Significant mediator: contributes materially to manifestation or interpretation but is not the dominant pathway.
Small	Supporting mediator: secondary, localized, or corroborative relation.

Node size must not be determined by word frequency alone. The application should combine causal centrality, recurrence across important scenes, perceptual necessity, climactic relevance, and narratorial emphasis. The model proposes an importance value; the application's own rule-based logic maps it to one of the three approved tiers.

9.3 Edge Requirements

Element	Requirement
Core relations	manifests through; appears in; speaks through; moves through; recorded by; perceived by; heard by; felt by; interpreted by; reported to; affects; reveals; possesses or animates
Solid edge	Explicit textual relation supported directly by a passage.
Dashed edge	Inferred, disputed, indirect, or ontologically ambiguous relation.
Line weight	May reflect narrative strength or recurrence, but the basis must be documented and accessible.
Direction	Edges must be directed where the relation is directional, such as ghost → portrait → seer.

9.4 Interaction Requirements

ID	Requirement	Specification	Priority
MAP-001	Hover / tap	Reveal role, relation, confidence, importance tier, and evidence count.	Must
MAP-002	Node selection	Open a drawer with explanation, passages, confidence, and alternative interpretation.	Must
MAP-003	Edge selection	Show relation label, explicit/inferred status, strength, rationale, and passage.	Must
MAP-004	Drag	Allow visual repositioning without altering graph data.	Must
MAP-005	Zoom and pan	Support mouse, keyboard, and touch inspection.	Must
MAP-006	Seer focus	Display only the selected seer's manifestation pathway.	Should

ID	Requirement	Specification	Priority
MAP-007	Filters	Show/hide secondary mediators, inferred edges, testimony relations, direct perceptions, and material, spatial, and sensory categories.	Must
MAP-008	View mode	Switch between Evidence View and Presentation View.	Must
MAP-009	Reset	Restore the generated layout and default filters.	Must
MAP-010	Text alternative	Provide a complete linear description of nodes, relations, and evidence.	Must

9.5 User Correction

- Dispute this interpretation.
- Suggest a different relation.
- Change importance level.
- Remove from revised map.
- Add missing mediator.
- Mark or unmark as a ghost-seer.
- Request stronger supporting evidence.

A user correction must create a revised map layer while preserving the original model-generated map, the reason for change, the timestamp, and the affected nodes or edges.

9.6 Network Interpretation

The network must be followed by a concise interpretive paragraph describing the dominant manifestation pathways, primary and secondary seers, and the degree to which spectral agency is bodily, architectural, acoustic, atmospheric, documentary, economic, or technological. The paragraph must reflect the currently visible and approved network rather than a stale initial version.

10. User Experience and Visual Design

10.1 Design Direction

The application must not reproduce the conventional visual vocabulary of Gothic entertainment, and it must not rely on literal spectral iconography such as ghost silhouettes, fog, or apparition imagery. The design instead stages a legible tension between two registers that mirrors the product's own premise: an AI system proposing structure and confidence around a genre that is fundamentally interpretive and ungraspable.

The analytical register carries the interface: structured layout, grid alignment, deep-navy and graphite tones, legible tables and criterion breakdowns. This register communicates the application's attempt to impose rubric-based order on the genre.

The spectral register is symbolic and restrained rather than decorative. Its primary expression is a signature cursor interaction: as the user moves the pointer, a soft, low-opacity point of light trails and fades behind it, evoking a will-o'-the-wisp. The effect must remain subtle, must never obscure content, must respect reduced-motion preferences, and must be implemented as an enhancement layer that degrades gracefully to a plain cursor when motion is disabled or unsupported.

Beyond the cursor motif, spectrality may be suggested through soft halos around active nodes, brief luminous transitions when evidence is revealed, or gently interrupted lines in the network map, but never through fog overlays, ghost silhouettes, or other literal representations. The genre is evoked symbolically; it is not illustrated.

10.2 Elements to Avoid

- Blackletter or imitation Victorian display fonts.
- Predominantly black screens and blood-red accents.
- Skulls, ravens, cobwebs, gravestones, candles, ruined castles, or cartoon ghosts.
- Literal ghost silhouettes, fog overlays, or other illustrative depictions of apparitions; spectrality must be evoked symbolically (e.g. the cursor motif), not illustrated.
- Animated fog that obscures content or causes motion discomfort.
- Jump scares, horror sound effects, or game-like suspense mechanics.
- Decorative complexity that competes with evidence and network legibility.

10.3 Approved Palette

Color	Value	Function
Deep Navy	#17324D	Primary interface actions, headings, mediator nodes, analytical structure.
Emerald	#16806A	Ghost-seer nodes, active markers, selected states, revision actions.
Warm Paper	#F7F5EF	Primary background.
Mist Gray	#DDE3E6	Secondary panels, ghost node interior, subtle separation.
Graphite	#292D32	Body text instead of pure black; anchors the analytical register.
Wisp Amber	#D9A441	Cursor-trail glow (low opacity), evidence-reveal transitions, and other spectral-register accents. Used sparingly and never as a solid fill.
Wisp Teal	#2E9E8C	Secondary cursor-trail tone; the glow may drift between Wisp Amber and Wisp Teal to avoid a static, decorative feel.

Dark Mode uses the same five roles at inverted lightness so the analytical/spectral duality and brand identity carry over exactly; it is not a separate design, only a re-tuned version of the same palette.

Role	Light Mode	Dark Mode	Function
Background	#F7F5EF	#15181C	Primary page background (Warm Paper / Ink).
Panel	#DDE3E6	#1F242A	Secondary panels, ghost node interior (Mist Gray / Slate).
Body text	#292D32	#EDEAE2	Body copy (Graphite / Parchment).
Structure	#17324D	#8FB2D6	Headings, mediator nodes, analytical accents (Deep Navy / Navy Light).

Role	Light Mode	Dark Mode	Function
Interactive	#16806A	#2FBE9D	Ghost-seer nodes, selected states, revision actions (Emerald / Emerald Bright).
Wisp cursor	#D9A441 / #2E9E8C	#F0C878 / #4FC4AE	Cursor-trail glow, brightened slightly in Dark Mode to keep the same visual weight against a darker background.

10.4 Typography

Element	Specification
Primary typeface	Aptos Body. Web font stack: "Aptos", "Segoe UI", Arial, sans-serif. Do not bundle a proprietary font file unless distribution rights are confirmed.
Body text	13 pt, approximately 17.33 CSS pixels; line height 1.55–1.65; left aligned; not fully justified. Fixed at 13 pt (not responsive down to a smaller size) to preserve readability across breakpoints.
Reading width	Fluid width proportional to the viewport (e.g. min(90vw, 1100px)) rather than a fixed pixel column, so text fills the available line width at any screen size instead of wrapping mid-screen with unused space on both sides.
Sentence presentation	The one-sentence-per-display-line convention applies only to designated approved copy blocks (e.g. the About page's Approved Main Paragraph and Approved Technical Paragraph in 11.2 and 11.3). General explanatory prose elsewhere flows as ordinary paragraphs at the fluid reading width, wrapping naturally without forced per-sentence line breaks.
Headings	Aptos semibold or bold, in a consistent hierarchy; avoid all caps except compact brand marks.

10.5 Navigation and Page Structure

Primary navigation: Home | About | Sample Analysis | Analyze | My Analysis | Language | Theme toggle. The Home page carries minimal text and a single inviting icon (the will-o'-the-wisp cursor motif is a natural anchor for this) that leads the visitor into About. The theme toggle (see 10.7) sits at the far right, after the language selector. Results pages may also include a persistent New Analysis action.

Page	Purpose
Home	Minimal text; a single icon inviting the visitor to click through. Selecting it opens About.

Page	Purpose
Analyze	Input, metadata, analytical mode, source language, output language, and rubric selection.
Results	Classification, evidence, criteria, network, ontology analysis, comparisons, rubric gaps, AI conversation, and exports.
Sample Analysis	One complete annotated example using public-domain or licensed text.
About	Purpose, audience, workflow, outputs, limitations, technical construction, creator, contact, and usage guidance for multilingual output, the Relational Manifestation Map, and the AI research conversation.
My Analysis	List of the authenticated user's saved analyses with reopen access; rerun and rubric-version comparison; deletion; and re-access to prior exports.

10.6 Responsive Behavior

- Desktop: maximum content width around 1,320 pixels; evidence drawer may open beside the network.
- Tablet: network remains interactive; evidence appears below or as an overlay.
- Mobile: all sections stack; network uses full width; filters open in a bottom sheet; 13 pt equivalent text is preserved.
- Exports must be generated from a dedicated print layout rather than a screenshot of the responsive page.

10.7 Theme Toggle (Light / Dark Mode)

A theme toggle sits at the top right of the primary navigation, beside the language selector, on every page. Its icon is a slender line-art Gothic lancet arch (the pointed arch silhouette of a cathedral window) framing a small disc: a sun in Light Mode and a crescent moon in Dark Mode. The arch is an architectural reference, not a horror motif, and stays consistent with the prohibition on literal Gothic iconography in 10.2 — no candle, flame, or creature imagery is used.

Selecting the icon swaps the interface instantly between the Light Mode and Dark Mode palettes with no page reload. The transition may use a brief, reduced-motion-safe cross-fade; it must not animate the will-o’-the-wisp cursor trail differently between modes beyond the color swap already specified in 10.3.

ID	Requirement	Specification	Priority
THEME-001	Placement	Show the toggle at the top right of the primary navigation on every page, including Reader View and Research View.	Must
THEME-002	Default and detection	On first visit, follow the operating system or browser’s prefers-color-scheme setting; do not default to Light Mode regardless of system preference.	Must

ID	Requirement	Specification	Priority
THEME-003	Persistence	Remember the user’s explicit choice (once they have toggled it) across sessions and across pages, independent of system preference.	Must
THEME-004	Icon states	Lancet-arch frame with a sun disc in Light Mode and a crescent moon in Dark Mode; the icon itself never changes shape, only the disc inside it, so the arch reads as a stable brand mark.	Must
THEME-005	Accessible label	Expose an accessible name stating the action, e.g. “Switch to dark mode” or “Switch to light mode,” not just an icon with no label.	Must
THEME-006	Export consistency	The combined PDF report (13.2) is always generated using the Light Mode palette, regardless of the user’s on-screen theme, so exported and printed reports stay legible and consistent.	Must

11. About Page and Public Copy

11.1 Placement

About must be available in the primary navigation and footer and open as a dedicated page rather than a modal. The page uses Aptos Body at 13 pt, a warm-paper background, a fluid reading width proportional to the viewport, and one sentence beginning on each new display line within the Approved Main Paragraph and Approved Technical Paragraph.

11.2 Approved Main Paragraph

Ghost Story Lens was created to make the boundaries of the ghost-story genre visible, testable, and open to revision rather than to impose a fixed yes-or-no definition.

The platform is intended for students, instructors, and researchers who want to examine how literary texts represent postmortem presence, supernatural agency, mediation, perception, and the return of unresolved histories.

To use the application, upload a supported file or paste a story, select an analytical mode and output language, review the criterion-based classification, explore the Relational Manifestation Map, and use the research conversation to question or refine the interpretation.

Each analysis provides a 1–10 Ghost-Story Fit score, a separate confidence assessment, supporting and contradicting passages, supernatural-entity classification, criterion-level explanations, corpus comparisons, rubric-gap findings, and an interactive network of the ghost, its mediators, and its ghost-seers.

Users may download a single combined PDF report containing the analysis summary and the relational map.

Multilingual output separates three settings: the language of the source text, the language of the interface, and the language of the explanations and report. Quoted evidence always remains in the story's original language, with an optional translation shown beneath it, so translation never substitutes for the primary source.

The Relational Manifestation Map can be explored by hovering or tapping a node or edge to reveal its role, confidence, and supporting passages, by dragging nodes to reposition them without changing the underlying analysis, and by using the filters to show or hide categories such as inferred relations or secondary mediators. A text alternative describing every node, relation, and passage is available for users who prefer or require it.

The AI research conversation lets a user ask questions about the analysis, the network, or the rubric, and every response is labeled as an Explanation, an Alternative Reading, or a Proposed Revision. Proposed changes to the analysis, the network, or the rubric are never applied automatically; they require the user's explicit approval and are preserved alongside the original output rather than replacing it.

Because the application relies on a historically situated and versioned rubric initially calibrated on British ghost fiction from 1764 to 1913, its conclusions are interpretive arguments rather than definitive genre judgments.

Analyses of texts from other periods, languages, and cultural traditions should therefore be treated as comparative and exploratory, and users should verify quotations, textual extraction, and AI-generated claims against the original work before using the results in teaching, publication, or formal research.

11.3 Approved Technical Paragraph

Ghost Story Lens was conceptually designed and its product documentation was developed in ChatGPT with GPT-5.6 Thinking.

The production application is planned for development and deployment through Replit and will use GPT-5.6 Sol through the OpenAI API for structured textual evidence extraction, criterion evaluation, supernatural-entity classification, relational network construction, multilingual explanation, and interactive research dialogue.

The application's own rule-based logic (not the AI model), rather than the language model, will calculate the final weighted score, apply classification thresholds, assign network-node sizes, preserve rubric versions, validate organized outputs, and generate downloadable files.

Each saved analysis will record its model, prompt, rubric, corpus, and analysis versions so that its results can be examined and reproduced as closely as the technical environment permits.

11.4 Creator and Contact

Creator

Created by Jong-Keyong (Jake) Kim
Questions, corrections, and revision suggestions: tsbym00@gmail.com

The creator and contact line must appear on the About page, in the site footer, and in combined PDF metadata. On screen, the contact email address must be a functional mailto link so selecting it opens the user’s own email application with the address pre-filled.

12. Multilingual Requirements

12.1 Initial Languages

- English
- Korean
- Simplified Chinese
- Traditional Chinese
- Japanese
- German
- French
- Spanish
- Hindi

The product must use Hindi rather than “Indian” as a language label. Additional South Asian languages must be added as distinct languages rather than grouped under a national label.

12.2 Separate Language Settings

Setting	Meaning
Source-text language	Language in which the literary text is written.
Interface language	Language used for navigation, forms, tooltips, and controls.
Output language	Language used for explanations, labels, report prose, and AI dialogue.

12.3 Evidence and Translation

ID	Requirement	Specification	Priority
LANG-001	Original evidence	Quoted evidence remains in the original source language.	Must
LANG-002	Optional translation	Provide an optional translation directly beneath the original passage.	Must
LANG-003	Translation transparency	Indicate when an interpretation depends materially on translation or terminology.	Should
LANG-004	Alternative translation	Allow a user to request or enter an alternative translation.	Should
LANG-005	Cross-linguistic notice	For texts outside the calibration corpus, display the approved exploratory-analysis qualification.	Must
LANG-006	Network labels	Prefer source-text names and stable role labels; translate relation labels according to output language.	Must

13. Export Requirements

13.1 Export Menu

Selecting Download Result generates a single combined PDF report containing both the analysis summary and the relational map. There is no separate JPG export; the PDF is the only downloadable artifact.

Export	Filename Pattern
Combined Results Report as PDF	Ghost_Story_Lens_[Title]_Combined.pdf

13.2 Combined Results PDF

The combined report must integrate the analysis summary and the network map in a stable, multipage research document. It should not be a raster screenshot of the web page. This PDF is the only downloadable export the application produces; there is no separate JPG of the summary or the map.

Section	Required Content
Page 1: Classification Summary	Title, author, spectrum, score, category, confidence, entity type, concise conclusion, rubric, corpus calibration, analysis date.
Page 2: Evidence and Criteria	Strongest supporting evidence, strongest counterevidence, abbreviated criterion scores, explanations, and passage identifiers.
Page 3: Relational Map	High-resolution network with all visible nodes fit within the frame and interactive controls removed, legend, relation styles, and interpretive paragraph. Preserve currently approved filters and user-revised positions where feasible.
Additional pages	Generated when evidence or the map cannot remain legible in three pages; may include corpus comparisons, rubric gaps, or revision history.

13.3 Technical Export Standards

ID	Requirement	Specification	Priority
EXP-002	PDF quality	Vector text where possible; print-quality network rendering; no clipped nodes, quotations, or labels.	Must
EXP-003	Metadata	Include title, rubric, corpus, model, prompt, output language, analysis date, and creator/contact in the PDF.	Must
EXP-004	Page layout	Use readable page numbering, repeated title/header, and stable margins.	Must

ID	Requirement	Specification	Priority
EXP-005	Static notice	State that interactive evidence and relation inspection remain available only in the application.	Must
EXP-006	Current state	Exports reflect approved user revisions and currently visible network filters; the report notes when it differs from the original model output.	Must

14. Data Model

14.1 Core Entities

Entity	Key Fields
User	Authentication identifier, role, locale, privacy preferences, saved rubrics, and ownership of analyses.
Document	Original filename, extracted text, source language, metadata, checksum, completeness flags, and passage index.
Analysis	Document reference, mode, rubric, corpus, model, prompt, score, confidence, entity labels, timestamps, and status.
Criterion Result	Criterion ID, score, weight, evidence, counterevidence, rationale, and confidence.
Evidence Passage	Stable passage ID, text, location, extraction confidence, optional translation, and source-language metadata.
Network Node	Label, role, subtype, color class, size tier, narrative importance, confidence, and passage references.
Network Edge	Source, target, relation, explicit/inferred status, strength, confidence, and passage references.
Revision	Original value, revised value, reason, user, time, affected objects, and approval state.
Rubric	Version, scope, criteria, weights, bands, exclusions, calibration, creator, and modification history.
Corpus Text	Bibliographic metadata, inclusion status, entity labels, benchmark annotations, and comparison features.

Entity	Key Fields
Export	Type, analysis version, selected filters, generation status, storage path, and creation time.

14.2 Versioning Rule

Immutability rule

A saved analysis is an immutable snapshot of a document, model, prompt, rubric, corpus, and application version. User corrections create a revision layer or a new analysis version; they do not rewrite the historical record.

15. AI and Technical Architecture

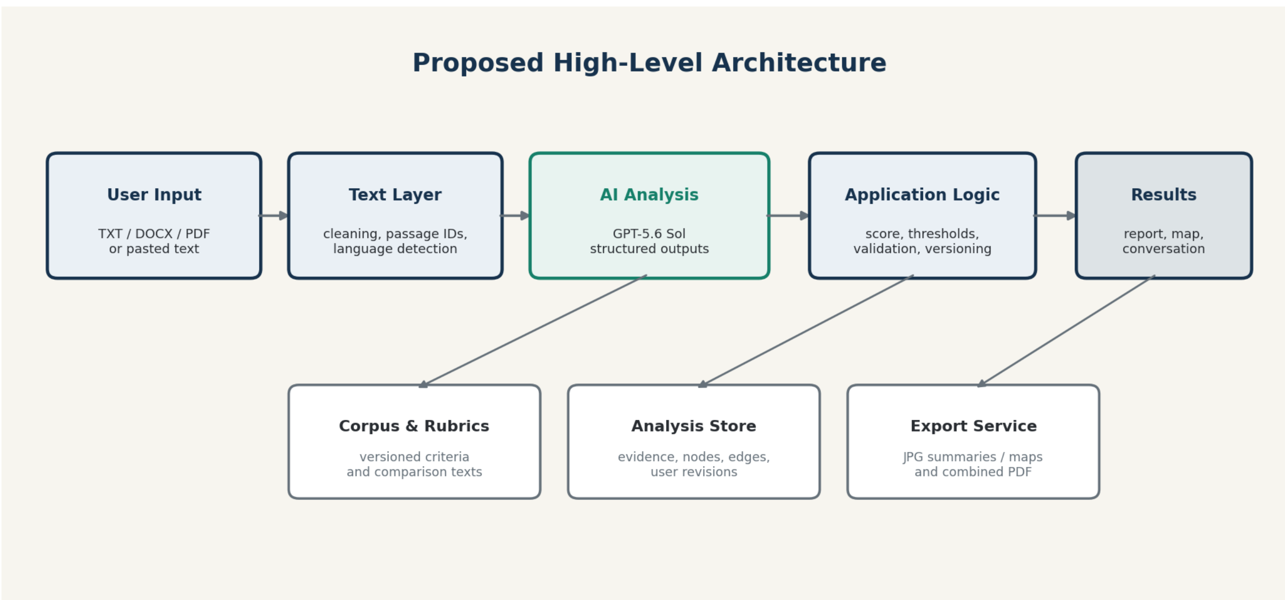

Figure 2. Proposed high-level architecture for the Replit implementation.

15.1 Implementation Environment

The production application will be developed and deployed through Replit. The exact framework may be selected during technical planning, but the PRD assumes a modern TypeScript web stack, internal, server-side endpoints, a relational database or equivalent persistent store, secure secret management, and a browser-based interactive graph library.

15.2 AI Model Responsibilities

- Extract entities, manifestations, mediators, seers, explanatory alternatives, and evidence passages.
- Evaluate criteria in organized output.
- Conduct the counterevidence pass.
- Propose nodes, edges, roles, and confidence for the relational map.
- Generate corpus-comparison explanations.
- Detect rubric gaps and draft revision proposals.
- Generate multilingual explanations and support the research conversation.

GPT-5.6 Sol is the specified production reasoning model. The integration should use the OpenAI API and organized outputs. Model identifiers and snapshots must be stored with every analysis so later model changes do not become invisible.

15.3 Application-Side (Non-AI) Responsibilities

- Calculate the final weighted score and classification band.
- Check that the AI's output follows the expected, consistently organized format.
- Match evidence to stored passages and reject unsupported references.
- Assign exactly three mediator-size tiers.
- Apply rubric and corpus versions.
- Store and compare original and user-revised results.
- Render network visualizations and exports.
- Enforce privacy, retention, permissions, and rate limits.
- Generate reproducibility metadata.

15.4 Suggested Service Boundaries

Boundary	Responsibility
Web client	Input, language selection, results, map interaction, conversation, revision controls, and export requests.
Application server	Authentication, file parsing, coordination, rule-based scoring, validation, persistence, and API security.
AI coordination	Versioned prompts, structured-output requests, retries, evidence validation, and model-response logging.
Corpus service	Reference texts, comparison features, benchmark annotations, and citation-safe metadata.
Graph service	Layout, filtering, revision overlays, text alternatives, and image rendering.
Export service	Multipage PDF generation from stable templates.

15.5 Reliability and Cost Controls

- Enforce maximum upload size and text length; disclose limits before submission.
- Segment long stories into stable passage windows but preserve global context and cross-passage relations.
- Cache immutable analysis results and exports.
- Use analysis jobs that can be safely retried or resumed without duplicating work.
- Retry format mismatches with bounded attempts and expose failure status rather than inventing partial results.
- Track token usage and per-analysis cost without exposing secret keys.
- Allow the product owner to change model or analysis depth through configuration rather than code edits.

16. Privacy, Copyright, and Research Integrity

16.1 Privacy

ID	Requirement	Specification	Priority
PRI-001	Private by default	Uploaded texts and reports are not public and are not added to the research corpus without explicit permission.	Must
PRI-002	Retention choice	Users may choose temporary analysis or saved analysis before processing.	Must
PRI-003	Deletion	Users can delete stored texts, reports, and exports they own.	Must
PRI-004	Secrets	OpenAI and other service credentials remain server-side in Replit secret storage or equivalent.	Must
PRI-005	Logs	Application logs must avoid storing full copyrighted texts unless necessary and disclosed.	Must

16.2 Copyright

- Do not publicly expose or redistribute full uploaded texts.
- Limit quotations in reports and exports to passages necessary for analysis.
- Require the user to confirm that they are authorized to upload and analyze the text.
- Prefer public-domain or licensed works for Sample Analysis and corpus demonstrations.
- Separate bibliographic metadata from text distribution rights.
- Provide a copyright notice clarifying that the application offers analysis, not permission to publish the underlying work.

16.3 Research Integrity

- Label AI-generated interpretations as AI-assisted and versioned.
- Preserve original and revised analyses.
- Display uncertainty and extraction warnings.
- Do not claim that statistical or model fit equals literary value, historical importance, or definitive genre identity.
- Record manual corrections and adjudication decisions.
- Provide enough metadata for a researcher to describe model, prompt, rubric, corpus, and date in a methods section.

17. Accessibility

ID	Requirement	Specification	Priority
A11Y-001	Contrast	Text, controls, and graph elements meet WCAG AA contrast requirements in both Light Mode and Dark Mode.	Must

ID	Requirement	Specification	Priority
A11Y-002	Color independence	No essential meaning is communicated by color alone; shape, ring count, line style, and text labels reinforce meaning.	Must
A11Y-003	Keyboard map	All interactive nodes, edges, filters, and drawers are reachable and operable by keyboard.	Must
A11Y-004	Screen readers	Nodes and edges expose meaningful accessible names, descriptions, roles, and evidence counts.	Must
A11Y-005	Text alternative	Provide a complete textual network description and ordered manifestation pathways.	Must
A11Y-006	Motion	Respect reduced-motion preferences and avoid unnecessary animation.	Must
A11Y-007	Scalable text	Support browser zoom and text enlargement without clipping or loss of controls.	Must
A11Y-008	Exports	PDF reading order, headings, alt text for the map, and tagged structure should be included where the export library permits.	Should

18. Validation and Quality Assurance

18.1 Human-Annotated Benchmark

Before public release, the default rubric and network extraction must be tested against a benchmark that contains unequivocal ghost stories, borderline cases, rationalized hauntings, non-ghost Gothic stories, demonic and occult narratives, metaphorical spectrality, multiple-seer stories, materially mediated ghosts, and stories without visible apparitions.

18.2 Evaluation Dimensions

- Primary and secondary entity identification.
- Overall Ghost-Story Fit band and rank consistency.
- Passage accuracy and quotation fidelity.
- Criterion-level agreement with human annotators.
- Counterevidence quality.
- Mediator and ghost-seer identification.
- Network-edge validity and direction.
- Narrative-importance size tier.
- Corpus-comparison usefulness.
- Rubric-gap usefulness and avoidance of overgeneralization.
- Multilingual output quality and preservation of original evidence.
- User-rated clarity and scholarly usefulness.

18.3 Test Categories

Category	Coverage
Unit tests	Score calculation, thresholds, size tiers, file validation, passage matching, filenames, and versioning.
Format tests	All AI responses validate against required structures; invalid responses fail safely.
Golden-set tests	Known benchmark texts produce expected entity, evidence, and network patterns within documented tolerances.
Stress and edge-case tests	Metaphorical "ghost" vocabulary, prompt injection inside literary text, malformed files, and contradictory narrators.
Accessibility tests	Keyboard, screen reader, contrast, zoom, reduced motion, and text alternatives.
Export tests	PDF completeness, clipping, resolution, metadata, Unicode, and multilingual typography.
Cross-browser tests	Current Chrome, Safari, Firefox, and Edge on desktop and mobile widths.
Load tests	Concurrent analysis jobs, long texts, export generation, and database limits.

18.4 Minimum Quality Gates

- No evidence quotation may be absent from the stored extracted text.
- No final score may be generated from missing required criterion results.
- Every displayed node and edge must have either evidence or an explicit "inferred" status with rationale.
- The network must remain readable at default zoom for the supported node limit.
- Combined PDF export must pass visual inspection for all benchmark languages.
- Critical privacy, deletion, authentication, and secret-management tests must pass before public deployment.

19. Analytics and Success Metrics

19.1 Product Metrics

Metric	Definition
Completion rate	Percentage of initiated analyses that reach a complete results page.

Metric	Definition
Evidence inspection	Percentage of users who open at least one supporting passage, counterevidence passage, node, or edge.
Interpretive engagement	Percentage of analyses that produce at least one research-conversation question or map correction.
Export use	Percentage of complete analyses exported as a combined PDF.
Return use	Authenticated users who complete another analysis within thirty days.
Failure rate	Extraction, model, output-format, passage-validation, graph-rendering, and export failures.
User rating	Perceived clarity, trustworthiness, usefulness, and accessibility.

19.2 Research Metrics

- Human–model agreement by criterion and entity category.
- Passage precision: proportion of cited passages judged genuinely relevant.
- Network node and edge precision/recall against annotated cases.
- Rate of user-approved rubric-gap proposals.
- Frequency and type of user corrections.
- Performance differences by period, genre, source language, and text quality.
- Stability of results across prompt or model updates.

Analytics must not store full literary texts merely for product measurement. Event data should use analysis identifiers and minimal metadata consistent with the privacy policy.

20. Risks and Mitigations

ID / Risk	Description	Severity	Mitigation
R-01	AI fabricates or mislocates evidence	High	Stable passage IDs, exact matching, format-checking, and rejection of unsupported citations.
R-02	Rubric appears universal	High	Persistent calibration language, Comparative Mode qualification, version display, and About-page limitation statement.
R-03	Score is mistaken for probability	High	Use “Fit,” prohibit probability language, and separate confidence from score.
R-04	Network becomes decorative or misleading	High	Require typed relations, evidence per relation, textual alternative, and user correction.
R-05	Long texts exceed cost or context limits	Medium	Text-length limits, segmentation, staged extraction, caching, and configurable analysis depth.

ID / Risk	Description	Severity	Mitigation
R-06	Copyrighted texts are redistributed	High	Private-by-default storage, limited quotations, no public full-text access, and rights confirmation.
R-07	Multilingual evidence is mistranslated	Medium	Preserve originals, optional translations, translation warnings, and language-specific evaluation.
R-08	Model update changes results silently	High	Store model snapshots and versions; benchmark before default model changes.
R-09	Aptos unavailable in browser	Low	Use approved fallback stack and do not distribute font files without permission.
R-10	PDF export clips complex networks	Medium	Dedicated export layout, node-limit handling, fit-to-frame validation, and automated visual tests.
R-11	Prompt injection inside uploaded fiction	High	Treat text as untrusted data, isolate instructions, use structured prompts, and restrict tool access.

21. Dependencies and Open Decisions

21.1 Dependencies

- A cleaned, versioned ghost-story corpus and contrast set with bibliographic metadata and legally appropriate access controls.
- A human annotation manual for rubric criteria, entity categories, mediators, seers, and edge relations.
- OpenAI API access to GPT-5.6 Sol and structured-output support.
- Replit project, deployment configuration, database, and secret management.
- A graph visualization library that supports accessible custom nodes, drag, zoom, filters, and export.
- A server-side PDF and image generation solution that handles multilingual text and high-resolution graphs.
- Language-specific reviewers or evaluation sets for the initial multilingual release.

21.2 Open Product Decisions

- Maximum accepted file size and token or word limit for one analysis.
- Whether multiple distinct ghosts are supported in the first public release or gated behind Research View.
- Authentication requirements for unsaved guest analyses (guests are now known to be blocked from Custom Rubric Mode and My Analysis persistence per FR-024; the open question is only how sign-in works for everything else, e.g. email/password vs. OAuth).
- Retention duration for temporary analyses and generated exports.
- Maximum default number of displayed network nodes and rules for collapsing low-importance nodes.
- Initial corpus-comparison retrieval method and number of comparison candidates evaluated before selecting three to five.
- Whether users may manually edit evidence quotations or only add notes and alternative interpretations.
- Whether the combined PDF includes full criterion detail by default or an abbreviated report with an optional extended mode.
- Benchmark size and minimum agreement thresholds required for public release.

22. Release Acceptance Criteria

1. A user can paste or upload a supported text and receive a complete report without exposing API credentials.
2. Every criterion contains passage-linked evidence and separately identified counterevidence or an explicit statement that none was found.
3. The final score is calculated by the application's own rule-based logic (not the AI) and displayed as fit rather than probability.
4. The result displays rubric, corpus calibration, model, prompt, and analysis versions in Research View.
5. The entity classifier does not rely solely on occurrences of ghost, spirit, specter, or apparition.
6. The interactive map contains a ghost or apparent-entity node when warranted, deep-navy mediator nodes in exactly three sizes, and emerald double-ring named or role-labeled ghost-seer nodes.
7. Every node and edge can reveal evidence, confidence, and interpretive rationale.
8. Users can filter, reposition, dispute, and revise the map without erasing the original output.
9. Reader View and Research View are both functional and accessible.
10. The visual design uses the approved restrained palette and avoids stereotypical Gothic imagery.
11. Body copy in the web application uses Aptos Body at 13 pt with approved fallbacks, and explanatory sentences begin on separate display lines where specified.
12. About is accessible from header and footer and contains approved purpose, workflow, outputs, limitations, technical construction, creator, and contact copy.
13. Interface and output support English, Korean, Simplified Chinese, Traditional Chinese, Japanese, German, French, Spanish, and Hindi.
14. Original-language evidence remains visible when the output language differs.
15. Users can download a single combined report as PDF containing the analysis summary and the network map.
16. The combined PDF includes classification, evidence, criteria, high-resolution network, legend, interpretation, metadata, creator, contact, and static-report notice.
17. Uploaded texts are private by default and users can delete stored analyses they own.
18. The benchmark, accessibility, export, privacy, and critical reliability tests pass the release thresholds set by the product owner.
19. A signed-in, authorized user can open the Custom Rubric Editor, edit criteria, weights, and score bands, and save only when weights sum to 100% and bands are monotonic and non-overlapping; the save creates a new rubric version, requires explicit approval, and never overwrites the default rubric.

Appendix A. Initial Analytical Rubric

The following weights and bands are initial product requirements, not immutable findings. They must be validated against human annotation and may change only through a documented rubric version.

Criterion	Initial Weight	Definition
Postmortem Identity	20%	Whether the entity is identified or strongly implied to have been a living human who died.
Manifestation	15%	How the entity becomes perceptible through sight, sound, touch, atmosphere, dreams, writing, objects, architecture, technology, or traces.
Causal Agency	15%	Whether the ghost communicates, compels, reveals, punishes, protects, alters bodies or property, or determines events.
Narrative Centrality	15%	Whether the principal conflict and resolution depend on the ghost or haunting.
Structure of Return	10%	Whether unresolved history, debt, violence, inheritance, burial, testimony, or mourning returns through the haunting.
Ontological Status	10%	Whether the apparition is confirmed, provisional, ambiguous, psychological, fraudulent, naturalized, or incompletely rationalized.
Mediation and Perception	10%	How ghosts, mediators, seers, narrators, investigators, and recipients of testimony form a manifestation pathway.
Generic Function	5%	What spectrality does: terror, mourning, testimony, ethical correction, comedy, religion, uncertainty, or social critique.

Score Bands

Score	Label	Interpretation
1.0–2.4	**Not a Ghost Story**	Little or no narratively consequential postmortem presence.
2.5–4.4	**Spectral-Adjacent Narrative**	Ghostly imagery, atmosphere, or apparitions without a central human ghost.
4.5–6.4	**Borderline or Ambiguous Ghost Story**	Substantial spectral evidence complicated by ambiguity, rationalization, or another supernatural category.
6.5–8.4	**Strong Ghost Story**	A deceased human presence substantially shapes plot, setting, evidence, or causation.
8.5–10.0	**Paradigmatic Ghost Story**	A central and causally effective ghost organizes conflict, historical return, and resolution.

Entity Classification Labels

- human ghost or revenant; ancestral ghost; ambiguous apparition; nonhuman spirit; demon or infernal entity; elemental or occult agency;
- vampire; werewolf; fairy; double or doppelgänger; psychic projection; dream or hallucination;
- fraudulent supernatural appearance; technological trace; metaphorical specter; unexplained phenomenon.

Appendix B. Network Ontology

Term	Operational Definition
Spectral source	Ghost, apparent ghost, revenant, apparition, or alternative supernatural entity under analysis.
Ambient mediator	Object, architecture, atmosphere, sensory condition, body, document, medium, technology, or person through which manifestation or interpretation occurs.
Attuned perceiver / ghost-seer	Character who sees, hears, feels, dreams, receives, records, or interprets the spectral phenomenon.
Direct perception	Spectral source is perceived without a distinct intermediate mediator in the passage.
Mediated perception	One or more material, spatial, sensory, textual, technological, or human mediators condition perception.
Testimony relation	A witness reports an experience to another character who did not directly perceive it.
Model-resistant relation	A case in which one entity crosses node categories or agency belongs to an ambient configuration rather than a discrete node.

Appendix C. Proposed Organized Outputs

The exact format will be finalized during engineering, but every AI stage should return validated data rather than free-form prose alone.

C.1 Analysis Object

```
analysis_id
document_id
rubric_id and version
corpus_id and version
model_id and snapshot
prompt_version
source_language / interface_language / output_language
criterion_results[]
```

overall_fit_score
classification_band
analytical_confidence
entity_classifications[]
supporting_findings[]
complicating_findings[]
network
corpus_comparisons[]
rubric_gap_proposals[]
created_at

C.2 Network Node

node_id
display_label
node_type
entity_subtype
color_class
size_tier
narrative_importance
confidence
evidence_passage_ids[]
alternative_interpretation
revision_status

C.3 Network Edge

edge_id
source_node_id
target_node_id
relation_type
explicit_or_inferred
strength
confidence
evidence_passage_ids[]
rationale
revision_status

Appendix D. Sources and Technical References

The PRD is based primarily on the approved product blueprint developed in this project and the dissertation's relational research architecture. The following entries document the principal internal and technical references.

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Final implementation note

This PRD defines the approved product behavior and design intent. Engineering choices may change when required by Replit, browser, graph-library, export-library, or API constraints, but any change that affects interpretive transparency, evidence grounding, rubric versioning, network semantics, privacy, accessibility, or required exports must be approved by the product owner.